Hazard Analysis
Software Engineering

Team #14, Troitsky Toolbox
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Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

This document follows the capstone Hazard Analysis template. Changes made to the template are listed in Table 3 (Section 8).

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[You are free to modify this template. —SS]

1 Introduction

1.1 Background

The Troitsky Toolbox is a web-based application that supports the McMaster Troitsky club, a student organization whose teams design, analyze, and build bridges out of popsicle sticks, white glue, and dental floss for the annual Troitsky Bridge Building Competition. The Toolbox centralizes the club's knowledge, including past designs, material testing data, and exit documents, and provides technical tools that allow teams to visualize their bridge models, estimate their performance, and check them against the competition rules. The problem and goals of the project are described in more detail in the Problem Statement and Goals document, and the system itself is specified in the Software Requirements Specification (SRS).

1.2 Hazard Definition

A hazard is defined as "a system state or set of conditions that, together with a particular set of worst-case environmental conditions, will lead to a loss" [**?**, p. 17]. For the Troitsky Toolbox, a hazard is therefore any condition of the system that, combined with how members use it, could lead to the loss of club knowledge, the disclosure or corruption of club data, the disruption of club activities, or members acting on incorrect information.

The Toolbox does not control any physical equipment, so the hazards it presents are primarily to the club's data and decisions rather than to people. The one exception is indirect: the Performance Analysis System produces estimates of a bridge's failure load, and a team that trusts an incorrect estimate may build or load test a bridge on that basis. This harm would be seen primarily in competition results, depending on the accuracy of the estimates. This document identifies these hazards, the failure modes that cause them, and the safety and security requirements that mitigate them.

2 Scope and Purpose of Hazard Analysis

The purpose of this hazard analysis is to identify the ways in which the Troitsky Toolbox could fail or be misused, assess their effects on the club and its members, and derive requirements that prevent or mitigate them. The resulting safety and security requirements are outlined in Section 6, and Section 7 states which of them will be implemented during the capstone, so that they are designed, implemented, and verified alongside the rest of the system.

The losses that could result from the hazards identified in this document are:

- **Loss of club knowledge:** Designs, test data, or exit documents are deleted, corrupted, or never captured, so that the knowledge the Toolbox exists to preserve is lost when members graduate. As the Toolbox becomes the club's central repository, a single failure can affect several years of accumulated information.
- **Loss of data integrity:** Stored data, such as material test results or competition rules, is incorrect or inconsistent, so that the results of the analysis tools are wrong.
- **Misleading analysis results:** A team relies on an incorrect performance estimate or compliance result, leading to a weaker bridge, a bridge that is disqualified for violating a competition rule, or wasted build time and materials.
- **Loss of confidentiality:** Members' personal information or the club's unpublished designs are disclosed to unauthorized users, including other competing universities.
- **Loss of availability:** The Toolbox, or one of its tools, is unavailable when members need it, such as in the weeks leading up to the competition.
- **Loss of trust:** Repeated failures lead members to stop using the Toolbox and return to scattered, undocumented information sharing, defeating the purpose of the project.

The analysis covers every software component of the Toolbox listed in Section 3, and the interfaces between the Toolbox and the external components it depends on, and the processes used to develop, deploy, and maintain the Toolbox, including its handover to future software subteam members. The following are outside the scope of this analysis:

- The physical construction and load testing of bridges, including the safety procedures followed by the club during testing.
- The CAD software that teams use to create their bridge models.
- The Troitsky Bridge Building Competition itself, including its rules and judging.
- Failures of the hosting infrastructure's physical hardware, beyond their effect on the availability and durability of the Toolbox's data.

3 System Boundaries and Components

The Troitsky Toolbox is structured around a central broker: each tool, as well as the shared data storage, is a service that registers with the broker, and all communication between services passes through it. The components below are

those described in Section S.1 of the SRS, and are shown in Figure 1. The requirements each component is responsible for are listed in Section S.2 of the SRS, using the prefixes given below.

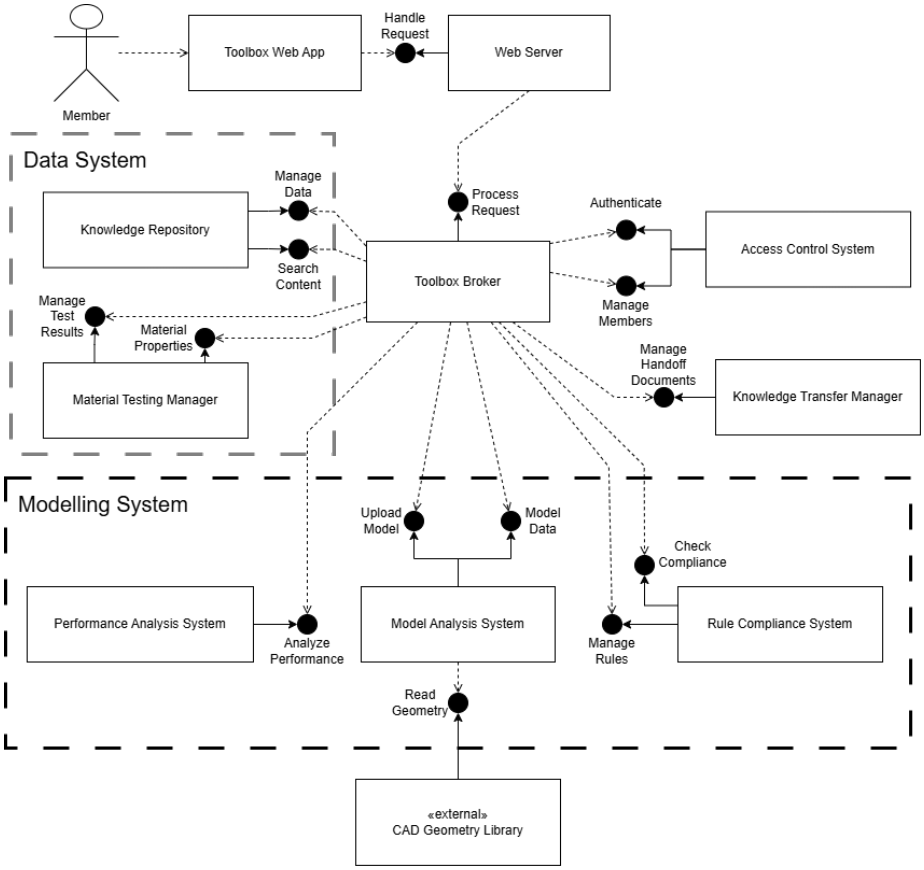


Figure 1: Components of the Troitsky Toolbox and the interfaces through which they interact.

3.1 Inside the System Boundary

The following components are developed by the team and are within the system's control and responsibility:

- **Web Server (WS):** Serves the Toolbox Web App and the public landing page, handles member login, and forwards requests for club services to the Toolbox Broker.
- **Access Control System (AC):** Manages member accounts and roles,

authenticates members, and decides which services and data each role may access.

- **Toolbox Broker (BR):** Registers services, verifies every request with the Access Control System, and routes it to the service that provides the requested operation. Because all communication passes through the broker, a failure of the broker affects every tool.
- **Knowledge Repository (KR):** Stores all club information, including documents, bridge models, material test results, competition rules, and handoff documents, and allows members to search and contribute to it.
- **Model Analysis System (MA):** Imports bridge models in the STEP format, renders them in 3D, and extracts the bridge properties needed for analysis, such as its dimensions and mass.
  - **Performance Analysis System (PA):** Estimates the failure load and load-to-mass ratio of a bridge model, using the extracted model data and the material properties provided by the Material Testing Manager.
  - **Rule Compliance System (RC):** Checks a bridge model against the competition rules configured by club members.
- **Material Testing Manager (MT):** Imports and stores material test results from CSV and Excel files, compares them against previous years, and derives the material properties used by the Performance Analysis System.
- **Knowledge Transfer Manager (KT):** Collects exit documents and role records from executives and team leads, and stores them in the Knowledge Repository.

3.2 Outside the System Boundary

The following entities interact with the system, but are not under its direct control:

- **Club members:** The general members, executives, and software subteam maintainers who use the Toolbox. Their decisions, including how they act on the results of the analysis tools, are outside the system's control.
- **CAD software and STEP files:** The software teams use to design their bridges, and the STEP files it exports. The Toolbox receives these files but does not control how they are created.
- **CAD/geometry library:** The external library used by the Model Analysis System to read STEP files.

- **Static model analysis APIs:** The external APIs used to perform the static model analysis of bridge model STEP files (SRS E.2).
- **Material test data files:** The CSV and Excel files produced by members during material testing, whose content depends on how the tests were performed and recorded.
- **Competition rules:** The official rules published by the competition organizers, which change from year to year and must be configured into the Rule Compliance System by members.
- **Hosting infrastructure:** The server, operating system, database server, and network on which the Toolbox is deployed.
- **Web browsers:** The browsers through which members access the Toolbox Web App.

4 Critical Assumptions

The following assumptions are made about the system and its environment. Each assumption states the class of hazards it removes from the scope of this analysis. Assumptions have been kept to a minimum, and failures that the system can reasonably detect or mitigate, such as invalid input files, service outages, and data entry errors, remain within the scope of the analysis.

1. **Analysis results are advisory:** Members treat the results of the Performance Analysis and Rule Compliance Systems as guidance, and the club verifies the final design against the official competition rules and follows its own safety procedures when load testing bridges. This assumption removes physical injury during bridge testing from the scope of this analysis. Hazards caused by incorrect or misleading results, and by results being presented in a way that encourages over-reliance, remain in scope.
2. **Members act in good faith:** Authenticated members do not intentionally misuse their access, for example by deliberately deleting or falsifying club data. This assumption removes deliberate insider sabotage from the scope of this analysis. Accidental misuse by members, and attacks by people outside the club, remain in scope.
3. **STEP files reflect the intended design:** The STEP files that teams upload accurately represent the bridge they intend to build. This assumption removes differences between a team's CAD model and its physical bridge from the scope of this analysis. Files that are invalid, corrupted, or that cannot be read remain in scope.
4. **The CAD/geometry library is correct:** The external library used by the Model Analysis System correctly reads the geometry of valid STEP files. This assumption removes defects within the library itself from the

scope of this analysis. Errors in how the Toolbox uses the library's output remain in scope.

5. **Test data reflects the tests performed:** The material test results that members import were measured using the club's testing procedure. This assumption removes errors in the physical testing process from the scope of this analysis. Data entry errors, malformed files, and physically impossible values remain in scope.

6. **Hosting infrastructure is maintained:** The server, operating system, and database server on which the Toolbox is deployed are kept updated and secured by whoever hosts them. This assumption removes vulnerabilities within the hosting platform itself from the scope of this analysis. Outages of the hosting infrastructure, and their effect on the availability and durability of the Toolbox's data, remain in scope.

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

Table 2: Failure Mode and Effect Analysis (FMEA)

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Web Server (WS) — SRS S.2.1							
Server down							H-WS-1
Server externally attacked							H-WS-2
Server overloaded from legitimate traffic							H-WS-3
Server loses connection to internet							H-WS-4
Server compromised							H-WS-5
Rogue actor gets access to change server							H-WS-6
Access Control System (AC) — SRS S.2.2							
Wrong user gets in							
Data leak (I/E)							H-AC-1
Password leak (I/E)							H-AC-2
Convenience leak (Superior gives inferior account access without clearance)							H-AC-3
Role category given extra rights							H-AC-4
Wrong person assigned to role category							H-AC-5
Right user locked out							
Forgot password							H-AC-6
Role category given less rights							H-AC-7
Right person not given role							H-AC-8

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Toolbox Broker (BR) — SRS S.2.3							
Knowledge Repository (KR) — SRS S.2.4							
Stored data lost or corrupted	Club data stored since the last backup is lost	Host or server failure, backup failure	H	L	Store backup of system (KR) off the server, on another platform	NFR-KR-2, SR-KR-1	H-KR-1
Item overwritten or deleted by mistake	A design or document is lost	Member error	M	M	Create confirmation prompts for deletions and modifications (e.g. replacing a file if uploading a copy), Access control protects file modification and deletion from all members	FR-KR-5, SR-KR-2	H-KR-2
Team's files visible to another team during the competition year	A team's unreleased design is exposed, contrary to SRS E.3	Access is checked by role only, not by which team owns the file	M	H	During the competition year, individual team files are marked as active and made available only to the owning team and executives	SR-KR-3	H-KR-3
Content cannot be found	Knowledge is lost in practice; work is duplicated	Missing or inconsistent tags, incorrect metadata	M	H	Require a title, a category from a fixed list, and a year on upload	FR-KR-3, SR-KR-4	H-KR-4
Repository unavailable	No saving or loading of stored data	Outage, storage capacity full	H	L	Tell the member when data was not saved	FR-BR-4, SR-KR-5	H-KR-5
Model Analysis System (MA) — SRS S.2.5							
Corruption of input file	A partly read or corrupted model returns wrong analysis results	Bad or interrupted CAD export	H	L	Show an import summary for the member to confirm before analysis or further processing	FR-MA-2, SR-MA-1	H-MA-1
Output display wrong, model analysis display not clear	Member misses a design flaw, bridge state misassessed	Wrong units or scale, errors in PA	M	M	Show units and a scale reference in the 3D view; check usability of the interface in the Winter CD — Usability Report	NFR-MA-2, SR-MA-2, Winter CD usability test	H-MA-2
Timeout	Member gets no response, either retries or gives up	Large model files, heavy load	M	M	Stop processing after MA_TIMEOUT and show error message	NFR-MA-1, SR-MA-3	H-MA-3

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Performance Analysis System (PA) — SRS S.2.5.1							
Calculation API error	No estimate, or a wrong ultimate load prediction	External analysis API unavailable, changed, or returning invalid values	H	M	Show an error instead of an estimate when the API fails or returns an invalid value; have test models to check new API calls against	NFR-PA-1, SR-PA-1	H-PA-1
Output confidence not understood	Estimate is misread or treated as a guaranteed result	Assumptions and limitations hidden, misunderstanding of service	M	L	Show the assumptions and limitations next to each estimate; check usability of the interface in the Winter CD — Usability Report	FR-PA-4, Winter CD usability test	H-PA-2
Rule Compliance System (RC) — SRS S.2.5.2							
Misinterpretation of rules (risk high for high penalty rules)	A non-compliant bridge passes the check and is penalized or disqualified at competition	Rule entered incorrectly, wrong year's rule set active, rule cannot be checked from the model, unit errors cause misrepresented comparison	H	L	Show the year of the rule set used, and highlight rules that require a manual check	FR-RC-5, NFR-RC-1, SR-RC-1	H-RC-1
Material Testing Manager (MT) — SRS S.2.6							
Input File Corruption	Failed or incorrect structural calculations	Information gets corrupted or dropped by the broker during upload	H	L	Given materials inputs, output values like stress or strain must fall within a range that is realistic for material type.	SR-MT-3	H-MT-1
Unit Mismatch	Incorrect Calculations during FEA, and incorrect display of materials properties	Uploaded data is in alternative units to what is used by the Model Analysis API	H	M	Have users select unit of their data during uploads and convert to appropriate units before application in API	SR-MT-1	H-MT-2
Grain Direction Neglection	The Performance Analysis System assumes incorrect material behaviour, predicting joints and other bridge pieces having much higher stress/strain tolerances than in reality	Wood has different material properties along the grain and perpendicular to the grain	H	L	Include a directional property for materials data that is saved as wood.	ST-MT-2	H-MT-3

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Knowledge Transfer Manager (KT) — SRS S.2.7							
Incomplete/Hollow Handoffs	Transition documentation contains broad details that future executives have little use for.	Executives treat transition documents as a "checkbox" to mark off and submit empty or vague transition documentation to complete the requirement.	L	M	Integrate the work documentation process as part of an executives week to week work, so when it come time to fill out transition documentation they will have a trove of data to pull from.	SR-KT-1 SR-KT-4	H-KT-1
Dangling & Broken References	Instructions reference external links that lead to 404 errors or missing technical context.	Documentation references external links or repository IDs that are later moved, renamed, or deleted in the Knowledge Repository.	M	L	References link to a look-up table that dynamically updates links to all documentation	SR-KT-2	H-KT-2
Knowledge Staleness	Members build bridges or perform testing using obsolete/disallowed specifications	Transition documents contain information that is only applicable to previous years rules or outdated tools	L	H	When opening old documentation, have a tab you can open that references the old rules and tools used compared to the current ones.	SR-KT-4	H-KT-3
Lack of Role Succession Continuity	New executives cannot locate their predecessor's records	An executive role is modified, split, or renamed between years, causing previous role records and onboarding material to become unassigned or orphaned.	H	L	Create Built in role modification, splitting, and inheritence so that records can be carried over to new or alternative roles	SR-KT-3	H-KT-4
Exposure of Sensitive Club Information	If stored as plain-text onboarding docs without field-level masking or strict role-based access, sensitive operational keys or personal data may leak	Handoff notes can contain sensitive credentials (e.g., third-party software licenses, passwords, personal phone numbers, sponsor contact details).	H	L	The Knowledge Transfer Manager must strictly forbid storing credentials as plain text in the KR	SR-KT-4	H-KT-5
Partial Transaction/Distributed State Failure	KT states that documentation exist within the KR when in fact it doesn't	When a transition document is completed the KT registers the completion but the handoff to the KR fails.	H	L	Have the KR send an acknowledgement	SR-KT-5	H-KT-6

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
System-Wide (SYS) — SRS S.2.8							
Compilation Error Due to Dependency Update	Project does not compile	A third part dependency updates and functions within that dependency that are used become degraded	H	L	Research and review dependencies Troitsky Toolbox will rely on and confirm that they are receiving quality updates and core functionalities Troitsky Toolbox depends on will continue to be functional.	SR-SYS-1	H-SYS-1
Code Deployed Without Testing or Review	Faults and Errors within the pushed product	Lack of testing and review	H	M	A CI/CD pipeline can help with some standardized testing along with a procedure outline for code reviews to alleviate some of the overhead in the reviews.	SR-SYS-2	H-SYS-2
Software Abandonment	Toolbox becomes unusable after initial capstone team leaves.	Initial capstone team leaves a project that is too difficult to maintain for the leftover software team.	H	M	Document every part of the development process, have multiple pieces of documentation for all critical system components, and train replacements before initial capstone team graduates.	SR-SYS-3	H-SYS-3

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

6.1 Functional Requirements

- **SR-KR-2:** The Knowledge Repository shall always prompt the user to confirm an action that will result in deletion or modification of stored files. [H-KR-2]
- **SR-KR-3:** The Knowledge Repository shall require that uploads of current competition year files are uploaded with a current team, limiting file access to only members of that team and executives. [H-KR-3]
- **SR-KR-4:** The Knowledge Repository shall reject an upload that does not include a title, a category selected from a standardized list, and a year. [H-KR-4]
- **SR-KR-5:** The Knowledge Repository shall return an error message whenever an uploaded file cannot be stored or a data retrieval error occurs. [H-KR-5]
- **SR-MA-1:** The Model Analysis System shall display a summary of inputted models to allow users to confirm the details and ensure accuracy and correctness. [H-MA-1]
- **SR-MA-2:** The Model Analysis System shall display the units of the inputted models, along with a scale reference in the 3D view. [H-MA-2]
- **SR-MA-3:** The Model Analysis System shall stop processing a model that has not finished within MA_TIMEOUT, and shall tell the member why processing stopped. [H-MA-3]
- **SR-PA-1:** The Performance Analysis System shall display an error message whenever the API fails, does not respond, or returns an inappropriate value. [H-PA-1]
- **SR-RC-1:** The Rule Compliance System shall display the competition year of the active rules with each result, and shall indicate when manual verification is required. [H-RC-1]

6.2 Non-Functional Requirements

- **SR-KR-1:** Backups of the Knowledge Repository shall be stored on an external platform. [H-KR-1]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Changes from Template

Table 3 lists the changes made to the structure of the capstone Hazard Analysis template and the reason for each.

Table 3: Changes from the capstone Hazard Analysis template

Change	**Reason**
Divided the Introduction into Background and Hazard Definition.	Separates the description of the project from the definition of a hazard that the rest of the document relies on.
Divided System Boundaries and Components into Inside and Outside the System Boundary.	Separates the components the team controls from the external entities whose failures the system can only detect or mitigate.
Listed new safety and security requirements only in this document, with the prefix SR.	Keeps each new requirement next to the hazards it mitigates; the FMEA table traces existing SRS requirements by their SRS IDs.
Added a References section.	Gives the source of the hazard definition.
Added Appendix — Changes from Template.	The HA checklist requires changes made to the template to be described in the document.

9 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

9.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

9.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

9.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

10 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Pedro Azevedo

1. In writing this deliverable, I believe that the team did well in the ideation and identification of the primary hazards present in the system and the development process. The team was able to separate the hazards based on their components strictly, so the splitting of work went well and was able to be smoothly integrated into the final deliverable. Since I have direct experience in the club I was able to provide direct insight into the hazards and the effects they would have on the club and its members.
2. The primary pain point during this deliverable was the linking and integration with the SRS. Since the SRS contains the FR and NFR that are referenced in the FMEA table, it was difficult to constantly ensure that referencing was up to date and correct, reflecting the changes to the SRS. This was resolved by having checkpoints where the team understood that the SRS was in a stable state. AI was also used to check and verify the references.

3. Of the listed risks, the primary ones the team had discussed were the risk of data loss and corruption, as the knowledge repository is central to the success of the service, and incorrect prediction or comprehension for the model analysis service, as this would directly affect teams' performance in competition. The other risks were identified during the writing of this deliverable mainly centred around operational hazards, such as the web server being down or overloaded, and the access control system being compromised. These were identified by the team by considering the state of the project once it is deployed and maintained by future members, outside the control of the current team.

4. Other than the risk of physical harm, 2 other types of risk in software projects may be data privacy risks and false confidence risks, which may present as future psychological harm or strict harm in user performance. These risks are important as they consider the effects of a software product that may not be immediately apparent, looking at what happens in the real world through the effect of the product. Having information leaked online, or having a user trust a result that is incorrect, both may lead to severe consequences for the user, outside the software domain.